

# Vittal Mani Bobbili

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## AI / ML Engineer

Experienced AI/ML Engineer skilled in developing scalable Python-based AI solutions and backend services supporting machine learning workflows. Demonstrated expertise in building real-time data pipelines, containerized deployments with Docker and Kubernetes, and integrating ML inference APIs with frontend applications. Proficient in computer vision, deep learning, and transfer learning with hands-on PyTorch experience. Strong problem-solving, collaboration, and analytical skills underpin a commitment to delivering high-quality, production-ready AI software.

## SKILLS

**Programming Languages:** Python, Java, TypeScript, JavaScript.

**Machine Learning Frameworks:** PyTorch, Transformer architectures, Large Language Models (LLMs), Generative AI, Reinforcement learning.

**Data Engineering and Infrastructure:** Distributed frameworks, Scalable infrastructure development, Data quality, Data generation, Inference time scaling.

**Software Development and APIs:** Generative programming, Programming models for interacting with models, FastAPI, REST APIs, Docker, Kubernetes.

## PROFESSIONAL EXPERIENCE

**AI / ML Engineer**, Zemoso Technologies

Jun 2024 - May 2025 | TG, IN

- Developed backend services and RESTful APIs using Python and FastAPI, implementing programming models for interacting with models and enhancing inference time scaling.
- Engineered real-time data processing pipelines with Kafka and PySpark, supporting scalable infrastructure development and distributed frameworks for AI research workflows.
- Containerized microservices with Docker and deployed on Kubernetes, ensuring scalable infrastructure and robust deployment for generative AI and ML workloads.
- Integrated REST APIs with React frontend, collaborating on generative programming features and facilitating communication of complex AI concepts across teams.
- Implemented machine learning algorithms including fine-tuning of Transformer architectures and Large Language Models (LLMs) using PyTorch to improve model performance and data quality.
- Contributed to generative AI research projects by developing training recipes and reinforcement learning workflows to optimize inference time scaling in multimodal models.
- Applied data generation techniques and ensured high data quality to support algorithm development and scalable infrastructure for advanced AI research.

## PROJECTS

### Optimized-YOLO: Live CCTV Road Accident Detection & Alert System

- Developed an AI-powered real-time detection pipeline using Python and YOLO to identify road accidents from CCTV footage with optimized inference latency.
- Implemented deep learning and OpenCV techniques for video frame processing and temporal analysis to improve accident event identification and reduce false alerts.

### AI-Based Crowd Management & Crime Prevention System – Smart India Hackathon

- Designed an AI surveillance solution using Python and machine learning to analyze crowd density and detect abnormal activities in real time from CCTV streams.
- Executed computer vision models and real-time data processing strategies to generate automated event alerts supporting security personnel in dynamic environments.

### Deep Learning-Based Vehicle Identification System for Hit-and-Run Investigation

- Implemented a multi-stage AI pipeline using PyTorch and transfer learning for vehicle detection and license plate recognition from surveillance videos.
- Developed image preprocessing and CNN-based classification methods to extract vehicle attributes, enhancing investigation workflows with structured vehicle information.

## EDUCATION

**Master of Science in Data Science and Artificial Intelligence**(CGPA: 3.83)

University of Central Missouri

May 2025 - May 2027 | M0, US

## **CERTIFICATIONS**

Introduction to Front-End Development – Coursera

Programming with JavaScript – Coursera

HTML and CSS in Depth – Coursera

Python for Data Science – NPTEL

## **publication**

Vindu- A Catering Services and Chef Hiring Mobile Application

Submitted to AIP Conference